

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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25ITR170

JAVA ASSIGNMENT

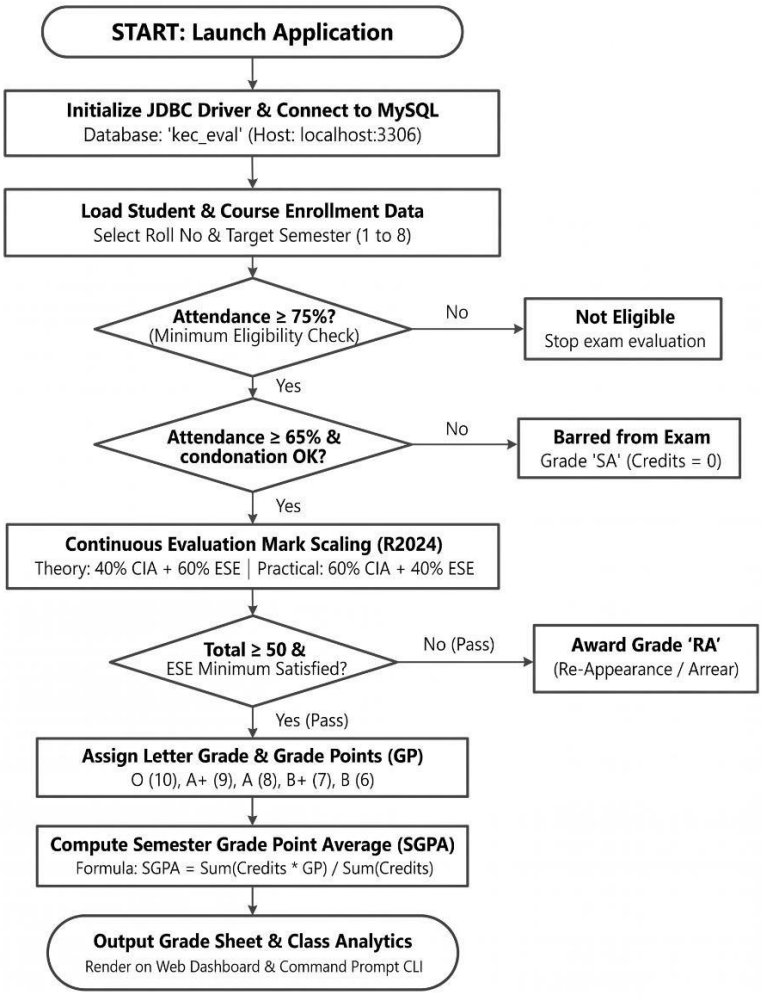
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE students;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
roll_no	varchar(20)	NO	PRI	*NULL*	
name	varchar(100)	NO		*NULL*	
department	varchar(20)	NO	MUL	*NULL*	
current_semester	int(11)	NO		*NULL*	
created_at	timestamp	NO		current_timestamp()	

2. COURSES TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE courses;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
course_code	varchar(20)	NO	PRI	*NULL*	
title	varchar(150)	NO		*NULL*	
credits	decimal(3,1)	NO		*NULL*	
course_type	enum('THEORY','PRACTICAL','EMBEDDED')	NO		*NULL*	
department	varchar(20)	NO		*NULL*	
semester	int(11)	NO		*NULL*	

3. ENROLLMENTS TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE enrollments;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	*NULL*	auto_increment
roll_no	varchar(20)	NO	MUL	*NULL*	
course_code	varchar(20)	NO	MUL	*NULL*	
semester	int(11)	NO		*NULL*	
classes_held	int(11)	NO		0	
classes_attended	int(11)	NO		0	
medical_condonation	tinyint(1)	NO		0	

4. ASSESSMENT MARKS TABLE:

Your SQL query has been executed successfully.

```
DESCRIBE assessment_marks;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
mark_id	int(11)	NO	PRI	*NULL*	auto_increment
enrollment_id	int(11)	NO	UNI	*NULL*	
cia_test1	decimal(5,2)	YES		*NULL*	
cia_test2	decimal(5,2)	YES		*NULL*	
assignment_quiz_lab	decimal(5,2)	YES		*NULL*	
ese_marks	decimal(5,2)	YES		*NULL*	

SOURCE CODE:

MAINAPP.JAVA:

```
package com.kec.eval.app;
import com.kec.eval.dao.*; import com.kec.eval.db.DBConnection; import com.kec.eval.model.*; import
java.util.*;
public class MainApp {
  static Scanner sc = new Scanner(System.in);
  static StudentDAO sDao = new StudentDAO();
  static AcademicDAO aDao = new AcademicDAO();
  public static void main(String[] a) {
    DBConnection.initializeDatabase();
    while (true) {
      System.out.println("\n=== KEC (R2024) EVALUATION SYSTEM ===");
      System.out.println("1.Add Student 2.Add Course 3.Enroll 4.Attendance 5.Marks (P1/2/3) 6.MarkSheet
7.Defaulters 8.Summary 9.Exit");
      System.out.print("Choice: ");
      switch (sc.nextLine().trim()) {
        case "1" -> addStudent();
        case "2" -> addCourse();
        case "3" -> enroll();
        case "4" -> attendance();
        case "5" -> marks();
        case "6" -> markSheet();
        case "7" -> defaulters();
        case "8" -> summary();
        case "9" -> { return; }
        default -> System.out.println("Invalid!");
      }
    }
  }
  static void addStudent() {
    System.out.print("RollNo Name Dept Sem: "); String[] p = sc.nextLine().trim().split("\\s+");
    System.out.println(sDao.addStudent(new Student(p[0], p[1], p[2], Integer.parseInt(p[3]))) ? "Added!" :
"Failed!");
  }
  static void addCourse() {
    System.out.print("Code Title Credits Type(THEORY/PRACTICAL/EMBEDDED) Dept Sem: "); String[] p =
sc.nextLine().trim().split("\\s+");
    System.out.println(aDao.addCourse(new Course(p[0], p[1], Double.parseDouble(p[2]),
CourseType.valueOf(p[3].toUpperCase()), p[4], Integer.parseInt(p[5]))) ? "Added!" : "Failed!");
  }
  static void enroll() {
```

```
    System.out.print("RollNo CourseCode Sem Held Attended: "); String[] p = sc.nextLine().trim().split("\\s+");
    int id = aDao.enrollStudent(p[0], p[1], Integer.parseInt(p[2]), Integer.parseInt(p[3]), Integer.parseInt(p[4]));
    System.out.println(id > 0 ? "Enrolled ID: " + id : "Failed!");
  }
  static void attendance() {
    System.out.print("EnrollID 1.Hours(Held Attn) 2.Condonation(Y/N Reason): "); String[] p =
 sc.nextLine().trim().split("\\s+", 3);
    int id = Integer.parseInt(p[0]);
    if (p[1].equals("1")) { String[] h = p[2].split("\\s+"); System.out.println(aDao.updateAttendance(id,
Integer.parseInt(h[0]), Integer.parseInt(h[1])) ? "Updated!" : "Failed!"); }
    else { String[] c = p[2].split("\\s+", 2); System.out.println(aDao.setMedicalCondonation(id,
c[0].equalsIgnoreCase("Y"), c.length > 1 ? c[1] : "") ? "Updated!" : "Failed!"); }
  }
  static void marks() {
    System.out.print("EnrollID Phase(1.CIA1 CIA2 | 2.Assig | 3.ESE/AB): "); String[] p =
sc.nextLine().trim().split("\\s+");
    int id = Integer.parseInt(p[0]);
    if (p[1].equals("1")) System.out.println(aDao.recordPhase1Marks(id, Double.parseDouble(p[2]),
Double.parseDouble(p[3])) ? "P1 Saved!" : "Failed!");
    else if (p[1].equals("2")) System.out.println(aDao.recordPhase2Marks(id, Double.parseDouble(p[2])) ? "P2
Saved!" : "Failed!");
    else if (p[1].equals("3")) {
      Enrollment e = aDao.getEnrollment(id);
      if (e != null && !e.isEligibleForESE()) { System.out.println("Barred from ESE (SA)! " +
e.getAttendancePercentage() + "%"); aDao.recalculateAndPersistEvaluation(id); return; }
      boolean ab = p[2].equalsIgnoreCase("AB");
      System.out.println(aDao.recordPhase3Marks(id, ab ? null : Double.parseDouble(p[2]), ab) ? "P3 Saved!" :
"Failed!");
    }
  }
  static void markSheet() {
    System.out.print("RollNo Sem: "); String[] p = sc.nextLine().trim().split("\\s+");
    StudentSemesterResult r = aDao.generateStudentSemesterMarkSheet(p[0], Integer.parseInt(p[1]));
    if (r == null || r.getStudent() == null) { System.out.println("Not found!"); return; }
```

```
    System.out.printf("--- GRADE SHEET: %s (%s) Sem:%s ---\n", r.getStudent().getName(), p[0], p[1]);

    System.out.printf("%-8s %-20s %-4s %-6s %-6s %-6s %-5s %-2s\n", "Course", "Title", "Cred", "Att%",
"CIA", "ESE", "Grade", "GP");

    for (CourseResult c : r.getCourseResults()) {

  String ese = c.isEseAbsent() ? "AB" : (c.getScaledEse() != null ? "" + c.getScaledEse() : "-");

      System.out.printf("%-8s %-20s %-4.1f %-5.1f%% %-6s %-6s %-5s %-2d\n", c.getCourseCode(),

c.getCourseTitle(), c.getCredits(), c.getAttendancePercentage(), c.getScaledCia(), ese, c.getLetterGrade(),

c.getGradePoint());
    }

    System.out.printf("Credits: %.1f | SGPA: %.2f | Percentage: %.2f%% | Result: %s\n",

r.getTotalEnrolledCredits(), r.getSgpa(), r.getEquivalentPercentage(), r.getSemesterStatus());

  }
  static void defaulters() {

    System.out.print("Semester (0 for All): "); int s = Integer.parseInt(sc.nextLine().trim());


    for (Enrollment e : aDao.getAttendanceDefaulters(s == 0 ? null : s))

      System.out.printf("%-10s %-15s %-8s %d/%d (%.1f%%) -> BARRED (SA)\n", e.getRollNo(),

e.getStudentName(), e.getCourseCode(), e.getClassesAttended(), e.getClassesHeld(),

e.getAttendancePercentage());
  }

  static void summary() {
    System.out.print("Semester: "); int sem = Integer.parseInt(sc.nextLine().trim());

    for (ClassSummary c : aDao.getClassPerformanceSummary(sem))

      System.out.printf("%-6s | Total:%-3d | Pass:%-3d | RA:%-2d | SA:%-2d | Pass:%.1f%% | Avg

SGPA:%.2f\n", c.getDepartment(), c.getTotalStudents(), c.getPassedEnrollments(), c.getRaEnrollments(),

c.getSaEnrollments(), c.getPassPercentage(), c.getAverageSgpa());

  }

}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;


/**

 * DBConnection - Database Connection Management for KEC Evaluation System

 * Kongu Engineering College (Autonomous) - Regulation 2024

 */

public class DBConnection {

    private static final String DEFAULT_HOST = "localhost";

    private static final String DEFAULT_PORT = "3306";

    private static final String DEFAULT_DB = "kec_eval";

    private static final String DEFAULT_USER = "root";

    private static final String DEFAULT_PASS = "0000";


    public static String getUrl() {

        String host = System.getenv("DB_HOST") != null ? System.getenv("DB_HOST") : DEFAULT_HOST;

        String port = System.getenv("DB_PORT") != null ? System.getenv("DB_PORT") : DEFAULT_PORT;

        String db = System.getenv("DB_NAME") != null ? System.getenv("DB_NAME") : DEFAULT_DB;

                        return "jdbc:mysql://" + host + ":" + port + "/" + db +
"?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";

    }


    public static String getUser() {

        return System.getenv("DB_USER") != null ? System.getenv("DB_USER") : DEFAULT_USER;

    }
```

```
    public static String getPassword() {
            return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") :
DEFAULT_PASS;
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found in classpath: " + e.getMessage());
        }
    }

    /**
     * Obtains a new database connection
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(getUrl(), getUser(), getPassword());
    }

    /**
     * Tests the database connectivity
     */
    public static boolean testConnection() {
        try (Connection conn = getConnection()) {
            return conn != null && !conn.isClosed();
        }
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
--------------------------------------------------------------------------------
                                MAIN MENU
--------------------------------------------------------------------------------
 1. Student Registration (Add New Student)
 2. Course Catalog Management (Add New Course: Theory/Practical/Embedded)
 3. Course Enrollment (Enroll Student in Course)
 4. Attendance Tracking & Medical Condonation (Update Hours & Approvals)
 5. Incremental Marks Entry ("Then and Now" - Phase 1, Phase 2, Phase 3)
 6. [REPORT] Student Semester Mark Sheet (SGPA, %, Letter Grades, Arrears)
 7. [REPORT] Attendance Defaulter Report (<65% & SA Flagged Students)
 8. [REPORT] Class Performance & Department Result Summary
 9. View System Records Overview (Students, Courses, Active Enrollments)
 10. Exit Application
--------------------------------------------------------------------------------
Enter your choice (1-10): 1
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
Enter your choice (1-10): 1

>>> 1. STUDENT REGISTRATION <<<
Enter Roll Number (e.g., 24ITR005): 25ITR170
Enter Student Full Name: VETRISELVAN K
Enter Department (e.g., IT, CSE, ECE, MECH, AIDS, EEE): IT
Enter Current Semester (1-8): 3

[SUCCESS] Student registered successfully!
Registered: 25ITR170 | VETRISELVAN K | Dept: IT | Semester: 3
```

2. COURSE CATALOG:

```
Enter your choice (1-10): 2

>>> 2. COURSE CATALOG - ADD COURSE <<<
Enter Course Code (e.g., 24ITT31, 24ITP32, 24ITE33): 24ITR11
Enter Course Title: Game Development
Enter Credits (e.g., 1.5, 3.0, 4.0): 3.0

Select Course Type (KEC R2024 Evaluation Category):
  1. THEORY     (CIA: 40% | ESE: 60% | ESE Min Threshold: 45% [27/60])
  2. PRACTICAL  (CIA: 60% | ESE: 40% | ESE Min Threshold: 50% [20/40])
  3. EMBEDDED   (CIA: 50% | ESE: 50% | ESE Min Threshold: 45% [22.5/50])
Enter Choice (1-3): 3
Enter Offering Department (e.g., IT, CSE, ECE): IT
Enter Offering Semester (1-8): 3

[SUCCESS] Course added to catalog successfully!
Added: [24ITR11] Game Development | Credits: 3.0 | Type: EMBEDDED | Dept: IT | Sem: 3
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
Enter your choice (1-10): 3

>>> 3. COURSE ENROLLMENT <<<
Enter Student Roll Number: 25ITR170
Student: VETRISELVAN K (IT, Current Sem: 3)
Enter Course Code to enroll: 24ITP32
Course: Object Oriented Programming with Java Lab (Type: PRACTICAL, Credits: 1.5)
Enter Enrollment Semester: 3
Enter Initial Classes Held (default 0): 0
Enter Initial Classes Attended (default 0): 0

[SUCCESS] Student successfully enrolled! (Enrollment ID: 19)
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 1
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter CIA Test 1 Marks (0-100): 92
Enter CIA Test 2 Marks (0-100): 93
[SUCCESS] Phase 1 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 2
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Assignment / Quiz / Lab Evaluation Marks (0-100): 90
[SUCCESS] Phase 2 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

3. PHASE 3 (POST SEMESTER):

```
Enter your choice (1-10): 5

>>> 5. INCREMENTAL MARKS ENTRY ("THEN AND NOW") <<<
Select Assessment Phase according to semester timeline:
  1. Phase 1 (Mid-Semester)       : Record CIA Test 1 (0-100) & CIA Test 2 (0-100)
  2. Phase 2 (End of Semester CIA): Record Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester ESE)  : Record End Semester Examination (ESE 0-100) or Mark Absent
  4. View Assessment Status of Student
Enter Choice (1-4): 25ITR170
Invalid numeric input. Please enter a valid number.
Enter Choice (1-4): 3

Enter Student Roll Number: 25ITR170
Enter Semester: 3

Select Course for Marks Entry:
  1. [24ITP32] Object Oriented Programming... (Type: PRACTICAL | Att: 92.3%)
  2. [24ITT31] Database Management Systems    (Type: THEORY    | Att: 100.0%)
Select Course (1-2): 2

--- [Phase 3: Post-Semester End Semester Examination (ESE)] ---
Course: [24ITT31] Database Management Systems (THEORY)
Was the student ABSENT in End Semester Exam? (Y/N): Y

[SUCCESS] Student marked as ABSENT (AB). Grade: AB, Grade Points: 0.
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
--------------------------------------------------------------------------------
Enter your choice (1-8): 6

Enter Semester to check (e.g., 3): 3

================================================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                         ATTENDANCE DEFAULTER & CONDONATION REPORT
                                  SEMESTER: 3
================================================================================================================
Roll No      Student Name         Dept     Course     Held     Attended Attd %     Condonation  Eligibility / Grade
----------------------------------------------------------------------------------------------------------------
24ADR010     Ananya S             AIDS     24ADT31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADC31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADL31    30       16          53.3% NOT APPROVED BARRED (Grade SA)
24CER003     Aravind K            CIVIL    24CET31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEC31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEL31    30       21          70.0% NOT APPROVED PENDING (Grade SA)
----------------------------------------------------------------------------------------------------------------
 Note: <65% Attendance: Shortage of Attendance (SA - Barred) | 65-74.9%: Requires Medical Condonation
================================================================================================================


Press [Enter] to return to the Main Menu...
```

STUDENT SEMESTER MARK SHEET:

```
Enter Student Roll Number (e.g., 24ITR001): 24ITR099
Enter Semester (e.g., 3): 3

=================================================================================================================
                                KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                                    PERUNDURAI, ERODE - 638060
                              OFFICE OF THE CONTROLLER OF EXAMINATIONS
                            GRADE SHEET - REGULATION 2024 (AUTONOMOUS)
=================================================================================================================
 Roll Number : 24ITR099          Student Name : Murasoli B                     Semester : 3
 Department  : IT                Regulation   : R2024 (KEC)                    Date     : 20-10-2026
-----------------------------------------------------------------------------------------------------------------
Course    Course Title               Type      Cr   Attd%    CIA1   CIA2   Asg    Sc.CIA   ESE(100) Sc.ESE  Total  Grd  Result
-----------------------------------------------------------------------------------------------------------------
24ITC31   Full Stack Development      EMBEDDED  4      96.7% 94.0   96.0   95.0      47.50 94.0     47.0     94.50 O    PASS
24ITL31   Programming in Java Labo...  PRACTICAL 1     100.0% 98.0   98.0   100.0     59.16 96.0     38.4     97.56 O    PASS
24ITT32   Programming in Java         THEORY    3      97.8% 90.0   92.0   94.0      36.76 90.0     54.0     90.76 O    PASS
-----------------------------------------------------------------------------------------------------------------
 Total Enrolled Credits : 8      |   Credits Earned : 8      |   Weighted Grade Points : 80.00
 SEMESTER GRADE POINT AVERAGE (SGPA) : 10.00 / 10.00
 EQUIVALENT PERCENTAGE               : 100.00%
 SEMESTER EVALUATION OUTCOME        : [PASSED] ALL REGISTERED COURSES CLEARED
=================================================================================================================
 Grading Standard (R2024): O(91-100)->10 | A+(81-90)->9 | A(71-80)->8 | B+(61-70)->7 | B(50-60)->6 | RA/SA/AB->0
 Rules: Theory (40 CIA/60 ESE, ESE Min 27/60) | Practical (60 CIA/40 ESE, ESE Min 20/40) | Pass Minimum: 50 Marks
=================================================================================================================


Press [Enter] to return to the Main Menu...
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
-------------------------------------------------------------------------
Enter your choice (1-10): 8

>>> 8. CLASS PERFORMANCE & DEPARTMENT RESULT SUMMARY <<<
Enter Semester: 3

=================================================================================================================
          KONGU ENGINEERING COLLEGE - SEMESTER 3 ACADEMIC PERFORMANCE SUMMARY
=================================================================================================================
Dept     | Students | Enrollments  | Passed   | RA(Arrear) | SA     | AB     | Pass %     | Avg SGPA
-----------------------------------------------------------------------------------------------------------------
CSE      | 1        | 1            | 1        | 0          | 0      | 0      | 100.00   % | 9.00
ECE      | 1        | 1            | 1        | 0          | 0      | 0      | 100.00   % | 8.00
IT       | 6        | 17           | 11       | 2          | 2      | 2      | 64.71    % | 5.47
-----------------------------------------------------------------------------------------------------------------
TOTAL    | 8        | 19           | 13       | 2          | 2      | 2      | 68.42    % | 7.49
=================================================================================================================


Press [Enter] to return to Main Menu...
```

DATABASE:

STUDENTS DATABASE:

roll_no	name	department	current_semester
24CSR015	Gokul R	CSE	3
24ECR008	Harini P	ECE	3
24ITR001	Aadhavan K	IT	3
24ITR002	Bhavani S	IT	3
24ITR003	Chandru M	IT	3
24ITR004	Deepika V	IT	3
24ITR099	Vijay Anand S	IT	3
25ITR168	Veeravigneshwar	IT	3
25ITR170	VETRISELVAN K	IT	3

COURSES DATABASE:

course_code	title	credits	course_type	department	semester
24CST31	Data Structures and Algorithms	3.0	THEORY	CSE	3
24ECT31	Digital Electronics & Logic Design	3.0	THEORY	ECE	3
24ITE33	Full Stack Web Development	4.0	EMBEDDED	IT	3
24ITE34	Cloud Computing and DevOps	4.0	EMBEDDED	IT	3
24ITP32	Object Oriented Programming with Java Lab	1.5	PRACTICAL	IT	3
24ITR11	Game Development	3.0	EMBEDDED	IT	3
24ITT31	Database Management Systems	3.0	THEORY	IT	3

ENROLLMENTS DATABASE:

	roll_no	course_code	semester	classes_held	classes_attended	medical_condonation	condonation_reason
1	24ITR001	24ITT31	3	50	46	0	*NULL*
2	24ITR001	24ITP32	3	30	28	0	*NULL*
3	24ITR001	24ITE33	3	60	54	0	*NULL*
4	24ITR002	24ITT31	3	50	43	0	*NULL*
5	24ITR002	24ITP32	3	30	21	1	Hospitalized with Dengue
6	24ITR002	24ITE33	3	60	49	0	*NULL*
7	24ITR003	24ITT31	3	50	28	0	*NULL*
8	24ITR003	24ITP32	3	30	20	0	*NULL*

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

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